Florian Julé

Oakland, CA | flojule@proton.me | flojule.github.io | US Permanent Resident

Mechanical and robotics engineer with 10 years of experiencedeveloping complex electromechanical systems from concept through production. Experienced in mechanical design, product development, systems integration, and robotics software.

Projects

Driver-Assisted Stair-Climbing Wheelchair Apr 2026 – Present

- Conceived and developed a 4WD articulated wheelchair for stair-climbing and off-road mobility, spanning mechanical design, ROS 2 software architecture (C++/Python), perception, and system integration on Jetson Orin Nano with a Gazebo digital twin.
- Architected and implemented the perception system from scratch across two cameras: an Intel RealSense D435i running YOLO26 semantic segmentation for general navigation and obstacle/stair detection, plus a D405 depth camera dedicated to stair navigation.

Cooperative Payload Transport with a Drone Swarm Jan – Mar 2026

- Drove trajectory generation for a drone swarm end to end: formulated and solved a CasADi OCP for collision-free, dynamically feasible paths, then deployed and validated the full pipeline on real hardware via ROS 2 and Crazyswarm2.

EKF SLAM on a TurtleBot3 Jan – Mar 2026

- Built a full SLAM stack from scratch in C++/ROS 2: 2D geometry library, simulator, odometry, and Extended Kalman Filter SLAM with unknown data association using LiDAR landmark detection.

Object Manipulation with Franka Arm Nov – Dec 2025

- Led perception and system integration for a 4-person team, delivering YOLO-based pick-and-place software for a Franka robotic arm on real hardware.

Work Experience

Joby Aviation **Santa Cruz, CA**

Staff Mechanical Engineer – Special Projects, Flight Research Nov 2023 – Feb 2025

- Owned mechanical system design, integration and test across novel aircraft programs in Joby's flight research division, spanning retractable landing gear, structural components, and cross-system integration.

Tail and Effectors IPT Lead Engineer – S4-2.1 production aircraft Oct 2018 – Nov 2023

- Cross-functional team lead for the development of tail, tilt nacelles, control surfaces, and propellers, advancing airframe designs towards FAA certification and production; managed 5 direct reports.
- Held end-to-end delivery responsibility for the first production-prototype aircraft and its conforming drawing package.
- Granted patent for the inboard nacelle tilt mechanism: Rotor assembly deployment mechanism.

Tilt Nacelles Lead Engineer – S4-2.0, pre-production aircraft Oct 2017 – Oct 2018

- Owned design, analysis, build, and test for the tilt nacelles on the Joby S4 pre-production aircraft.
- Served as flight test engineer throughout the flight test program.

Mechanical Engineer – S4-1.0, first full-scale eVTOL aircraft Feb 2017 – Oct 2017

- Supported composites manufacturing and structural testing for the first Joby S4 full-scale prototype.

Education

Northwestern University **Evanston, IL**

M.S. in Robotics Sep 2025 – Sep 2026

University of Michigan **Ann Arbor, MI**

M.S. in Aerospace Engineering Sep 2015 – Dec 2016

Arts & Métiers ParisTech **Metz/Paris, France**

B.S. in Mechanical & Industrial Engineering Sep 2013 – Jun 2015

Skills

Robotics	Sensor Integration (IMU, Cameras, LiDAR, Encoders), Localization (Kalman & Particle Filters), SLAM, Motion Planning, Control (PID, Feedforward)
Software	Python, C++, ROS 2, Git, Linux
Mechanical	Mechanism design, Composite structure, CAD (Catia, Onshape), FEA (Abaqus), GD&T (ASME Y14.5)
Languages	French, English